

A glass bottle is tilted, pouring a liquid into a body of water. The liquid creates concentric ripples on the water's surface. The bottle is in the upper left, and the ripples spread out towards the bottom right.

Reproducible Linux Installations

on Your Laptop Without Data Loss

...or having multiple Linux installs side-by-side

The Problem

The installer has a lot of capabilities, but it's *dangerous* if you don't know exactly what you are doing.

Modern OS installers are great at **first installs** on empty disks.

Most have an interactive partition editor,

but:

One wrong click and your data is gone - like that.

It's not the installer's fault. It's

- that humans make mistakes under pressure, at 2am, or
- simply because the layout wasn't obvious.

Many colleagues told me they *accidentally* wiped their data on (re)install.

Many reports on the internet where you can read

"I lost my data on (re)install, even when I tried hard to keep it"

The Strategy: Know Your Disk State

Two modes — decided by what's already on disk:

Situation	Layout
Empty disk / single OS planned	single-os
Existing OS(es) detected	multi-os

A Python pre-script inspects the real partition table before touching anything.

It aborts if the state doesn't match expectations. - No guessing. No silent data loss.

BUT:

Have you defined your disk layout correctly - otherwise **data loss**

TEST before you install one real hardware

!!! HAVE BACKUPS !!!

Disk Layout: Multi-OS Example

```
1  EFI                <- preserved, reused
2  MS Reserved        <- untouched
3  Windows            <- untouched
4  WinRecovery         <- untouched
----- startpartition 5 -----
5  /boot              <- created or emptied
6  OS                 <- created or emptied
   (LUKS, btrfs)
   subvol: root
   subvol: home
-----
7  data               <- never touched on OS install
   (LUKS, btrfs)       created or verified by Ansible
   subvol: holger      -> /home/holger
   subvol: images      -> /var/lib/libvirt/images

8  winexchange        -> exchange data between
   (NTFS, unencrypted) Windows and Linux
```

Rules

- Partitions 1–4: **never touched**
- Partitions 5+6: **created on first install, emptied on reinstall**
- Partitions after 6: **never touched**

Ansible creates or verifies the state of these partitions
- EFI is reused → other OSes keep booting
- Distribution swap on reinstall: possible

Ubuntu → Fedora on same partitions. Windows and data: untouched.

Partition Layout:

4 TB disk - P7 $\approx$ 3490 GiB

Fixed partitions total: 606 GiB. Scale: each '=' $\approx$ 82 GiB

Part	Size	Bar
1	2 GiB	=
2	256 MiB	=
3	200 GiB	==
4	2 GiB	=
5	2 GiB	=
6	200 GiB	==
7	~3490 GiB	=====
8	200 GiB	==

1TB disk, 2TB disk and 4TB disk in relation

1TB: |1|2| 3 (200G) |4|5| 6 (200G) | 7 (~418G) | 8 (200G) |

2TB: |1|2| 3G |4|5| 6G | 7 (~1442G) | 8G |

4TB: |1|2|3G|4|5|6G| 7 (~3490G) |8G|

The Builders (1)

Everything runs in **Podman containers**. No special host setup beyond: `podman` · `tar` · `rsync` - `gpg`

`faibuilder` / `ksbuilder`

faibuilder: produces a **bootable disk image** (btrfs, thumbdrive-ready) with the installer (FAI-based)

ksbuilder: produces a an **image** with a partition labeled "**OEMDRV**" which is found by the second connected image containing a standard Fedora/RHEL (and similar) ISO image.

Both builders add an additional partition labeled "**EXTRA**" for additional content:

- Ansible content (encrypted)
- Offline package repos (e.g. Fedora 43 WS — no Anaconda ISO exists, only LIVE ISO)
- Whatever you need

Installations run fully offline. No internet required.

The Builders (2)

Everything runs in **Podman containers**. No special host setup beyond: `podman` · `tar` · `rsync` - `gpg`

`nsbl dr` (Ansible builder)

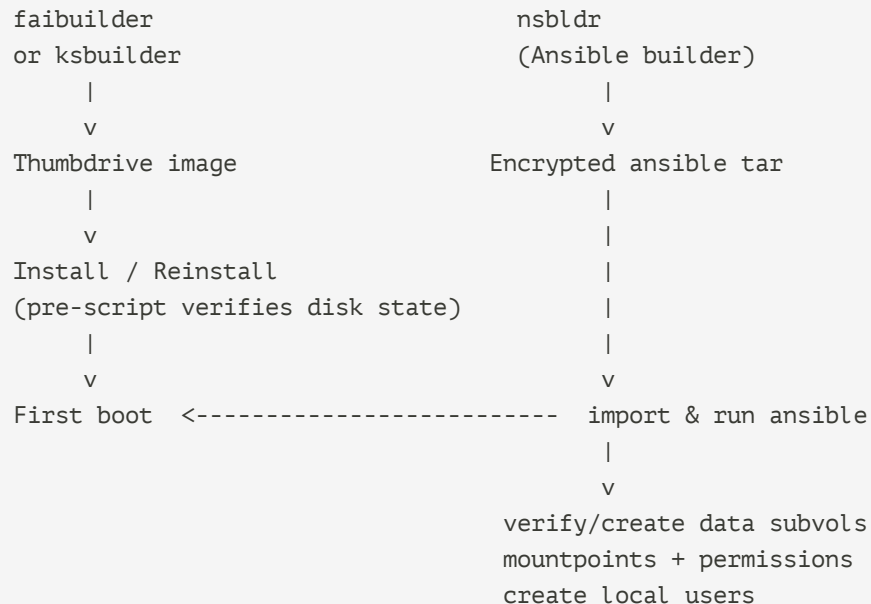
Packages Ansible content into a **GPG-symmetrically encrypted archive**.

Extracted after first boot → run via a dedicated system user (f.e. hoo).

Archive can contains:

- Ansible inventories
- Ansible files (files that should be delivered statically or via template)
- Ansible playbooks
- Ansible roles
- Ansible collections

The Full Workflow



Live Demo

First install → Reinstall with a Distribution swap

Windows and data: untouched throughout

Ask questions while it runs — installs take a few minutes

Takeaways & Links

Even if your setup looks nothing like mine — steal what's useful:

- LUKS + Btrfs subvolumes as a sane data/OS separation
- Pre-script state verification before any destructive action
- Offline-first installer images via Podman
- Agama support: **work in progress**

hoonetorg · github.com/hoonetorg

Slides + code links:

[Slides](#)

Source code:

Watch out for existing repos and changes at github.com/hoonetorg.

There will be a repo (not named yet) consolidating all required tools, containing examples.